

Leveraged Loan Index Futures

A Primer

- On September 14, market participants will be able to access the USD leveraged loan market through a new CME Credit Index Future. The future builds on the success of other futures available in the credit market referencing high grade and high yield bond indices as well as the established loan ETF and Total Return Swap products.
- Credit Index Futures provide a flexible and scalable instrument for managing credit exposures. The standardised nature of the instrument, along with it being exchange traded and centrally cleared, means that it has appeal to a broad array of market participants and can provide cost-effective access to corporate benchmarks.
- In this note, we explain how the new futures will work and compare and contrast them with the existing products available to market participants looking to gain loan market exposure. We look at the underlying composition of the reference index, along with that of the two largest ETFs and compare the returns and drawdown experience over the past ten years.
- Exchange data will be available on our DataQuery portal along with Fair Value, Premium-to-NAV and Excess Return indices. We will also include the product in our quarterly roll update and volume analysis.

Credit Derivative and Index Research

Saul Doctor ^{AC}
(44-20) 7134-1539
saul.doctor@jpmorgan.com
J.P. Morgan Securities plc

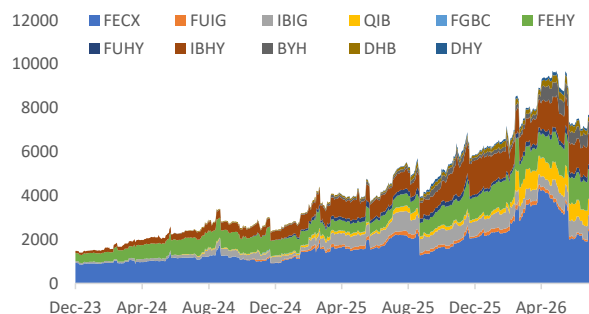
Pavan D Talreja, CFA
(1-212) 834-2051
pavan.talreja@jpmchase.com
J.P. Morgan Securities LLC

Dmytro Shelukhin, CFA
(44-20) 7134-2151
dmytro.shelukhin@jpmorgan.com
J.P. Morgan Securities plc

North America Corporate Credit - Distressed, High Yield and Leveraged Loans Strategy

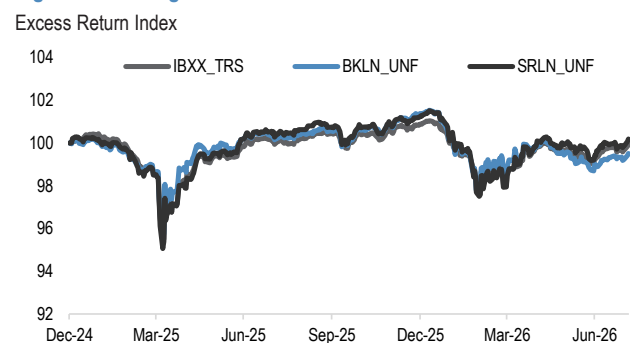
Nelson Jantzen, CFA ^{AC}
(1-212) 270-1169
nelson.r.jantzen@jpmorgan.com
J.P. Morgan Securities LLC

Figure 1: Credit Index Futures Open Interest



Source: J.P. Morgan.

Figure 2: Leveraged Loan TRS versus ETF Returns



Source: J.P. Morgan.

See page 21 for analyst certification and important disclosures.

Table Of Contents

A Brief Introduction to Leveraged Loans	3
Macro Access to the Leveraged Loan Market	6
Introducing Credit Index Futures	7
Benchmarks and ETFs	9
Composition	10
Returns and Correlations.	13
Contract mechanics.	16
Credit Futures Data in DataQuery	19
Quarterly Roll Notes	19

A Brief Introduction to Leveraged Loans

A leveraged loan is typically a senior, secured loan made to a company whose credit profile is below investment grade. These are typically used to finance M&A, LBOs, dividends/recapitalizations, and other general corporate purposes. The common structures include Revolving Credit Facility (RCF) for liquidity/working capital, the main institutional tranche Term Loan B (TLB), and sometimes additional tranches (Term Loan A, C, etc.).

A syndicated loan is structured, arranged, and administered by one or several commercial or investment banks, known as arrangers. Despite “bank loan” roots, most leveraged loans are now syndicated and distributed to retail and institutional investors (CLOs, Mutual Funds, ETFs, SMAs, Pensions, Insurers, Hedge funds, Private Wealth, HY Funds) rather than being held by relationship banks. Syndicated loans are less expensive and more efficient to administer than traditional bilateral—or individual—credit lines.

Most leveraged loans are floating-rate instruments with an all-in rate encompassing a base rate (SOFR) plus a credit spread. When issued, investors typically care about the OID (original issue discount) and call protection (often limited to six months) because loans are usually prepayable at Par. Since the coupon resets with the base rate, loans have low interest-rate duration and limited convexity. Institutional term loans usually have minimal required amortization (i.e. ~1% per year) and typically carry longer settlement.

A large portion of institutional loan demand (~70%) comes from CLOs (Collateralized Loan Obligations), which buy diversified pools of senior-secured loans and finance them via tranche liabilities (AAA down to equity). Key drivers of syndicated loan performance include credit fundamentals, base rates, CLO creation, inflows/outflows, documentation quality, refinancing risk, and industry cycles.

Importantly, leveraged loans are not securities and are not sold to the public unlike the more common high yield bonds. In Table 1, we highlight the main differences between leveraged loans and high yield bonds.

Table 1: Comparing Loans to Bonds

Dimension	Leveraged Loans	High-Yield Bonds
Seniority / security	Senior, often secured	Usually unsecured or second-lien; structurally subordinated to secured bank debt
Rate type	Floating (SOFR + spread)	Fixed-Rate Coupon
Investor base	CLOs, Mutual Funds, ETFs, SMAs, Pensions, Insurers, Hedge funds, Banks, Private Wealth, HY Funds	Mutual funds/ETFs, Insurers, Pensions, Equity and Income Funds, Investment-Grade Funds, Hedge Funds
Documentation	Credit agreement; Agented lender group	Indenture; Trustee structure
Covenant style	Historically maintenance; now often covenant-lite for TLBs	Mostly incurrence covenants
Maturity / amortization	5–7 years common; minimal required amortization (e.g., ~1% per year) and a bullet repayment at maturity	7–10 years common; bullet maturity
Call protection / refinancing	Often 6-month soft call (easier to refinance or reprice)	Stronger/clearer call protection (non-call periods, make-whole)
Trading & settlement	Less liquid; longer settlement; assignment mechanics	Generally more liquid; faster settlement; simpler trading rails
Recovery	Historically higher expected recovery due to secured seniority	Historically lower recovery due to unsecured nature
Why borrowers might prefer	Floating-rate funding, Speed and certainty, Amend-and-Extend/Repricing pathways, Flexible capital structure	Fixed-rate funding, Longer tenor, Clearer call structure and capital planning, Diversified investor base,
Why investors might prefer	Less duration or rate risk, seniority/collateral, expect rates to rise or stabilize (carry), lower volatility	Constructive view on rates falling, Liquidity, Upside beyond par (convexity)

Source: J.P. Morgan.

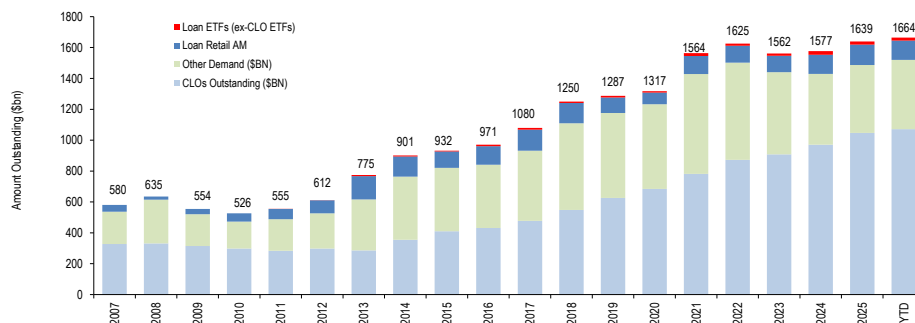
Characteristics of the Broadly Syndicated Leveraged Loan Universe

The institutional leveraged loan universe stands at \$1.66trn outstanding, which has grown \$25bn in 2026 (+2%) and \$103bn since YE23 (+7%) amid a rise in net new issuance, loan-for-bond takeouts, and a balanced mix of private and public market takeouts. For context, the loan market contracted by \$84bn (or 5%) from mid-2022–YE23 after expanding by 11% per annum over the preceding decade.

The largest buyer base by a wide margin is CLOs which hold \$1.07trn (or 64%) of the broadly syndicated loan universe. Retail holds an additional \$144bn (or 9%) of which a small sliver consists of the Loan ETF products (\$19bn). Lastly, the other component of demand such as institutional accounts (pension, insurance companies or SMAs), hedge funds, banks, private wealth management, and HY funds hold the balance of the \$449bn BSL (or 26%).

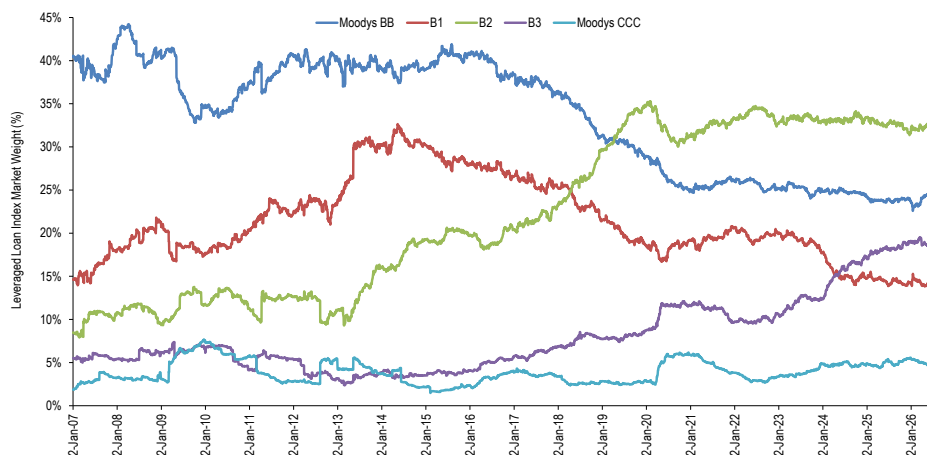
The ratings mix for the leveraged loan index weakened over the past decade amid a combination of aggressive issuance (LBOs) and weak agency trends (downgrades). For context, BB, B1, B2, B3, and CCC-rated loans account for 25%, 14%, 32%, 18%, and 5% of the Leveraged Loan index, respectively, which a decade ago stood at 40%, 27%, 19%, 5%, and 3%. Lastly, industry concentration in the leveraged loan product is relatively high. For context, the four largest industries including Technology (14%), Healthcare (12%), Services (11%), and Financials (10%) account for 47% of the leveraged loan index

Figure 3: The Leveraged Loan Market is \$1.6Trn in size with CLOs the largest buyer base



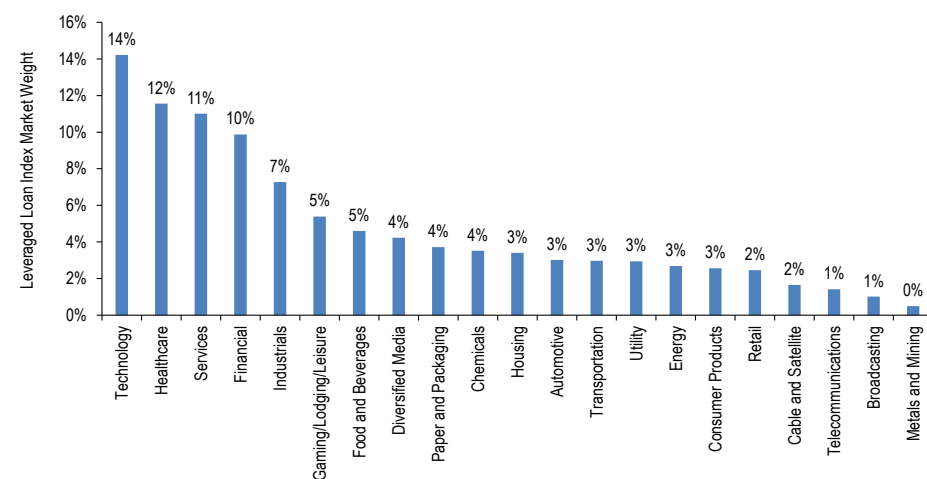
Source: J.P. Morgan; S&P/IHS Markit, Lipper FMI, Intex

Figure 4: The ratings for the Leveraged Loan index eroded over the past decade



Source: J.P. Morgan; S&P/IHS Markit

Figure 5: The Loan index is heavily weighted in four industries



Source: J.P. Morgan; S&P/IHS Markit

Macro Access to the Leveraged Loan Market

Investors looking to access the leveraged loan market through standardised macro products currently have two options: ETFs or Total Return Swaps (TRS) on an S&P benchmark indices. We estimate that macro products (ETFs and TRS) are now 15% of the daily USD loan volumes in the market (Figure 7).

Growth in both leveraged loans and the ETF market over the past few years means that the market currently stands at \$20bn with the largest two ETFs, BKLN and SRLN \$12bn combined. Leveraged loan ETF volumes have grown rapidly over the past few years and currently stand at \$572mn/day. ETFs have been a useful product for market participants to rapidly gain access to the leveraged loan market since accessing the underlying loans can be more difficult. Borrowing and shorting the ETFs has however been more difficult with the cost of borrow becoming very costly at some point. The difficulty in trading ETF versus the underlying portfolio of loans, where the borrow can be even more costly, has meant the ETF is primarily a vehicle for long risk expressions. A general introduction to Credit ETFs and their uses can be found in [Credit ETF Trading Strategies: Macro Views Made Easy](#).

Standardised Total Return Swaps on the S&P benchmark index have also grown sharply in liquidity over the past few years with 150mn/day of volume in the first half of 2026 (Figure 7). The benchmark underlying the indices is the S&P UBS USD Liquid Leveraged Loan Index (LLL) with the Bloomberg ticker IBXXLLTR which tracks a subset of the broader loan index and aims to represent the 100 largest, most liquid loans in the market. The nature of the TRS product means that the TRS price level has tended to trade at a discount to the benchmark index level reflecting hedging demand. That said, the product does enjoy two-way demand as many investors see it as an easy way to gain unfunded access to the leveraged loan market. An introduction to credit TRS can be found in [The iBoxx TRS Handbook: Trading Credit Total Return Swaps](#).

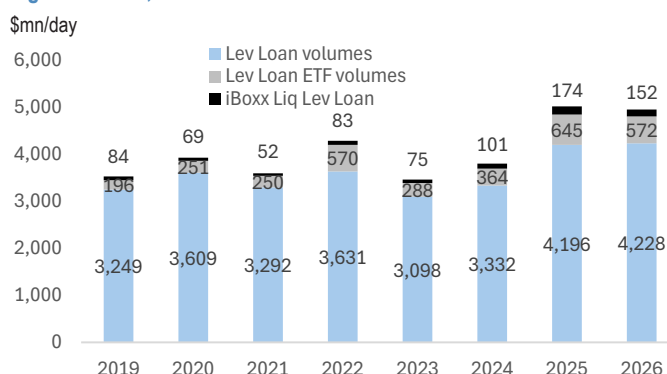
The new CME leveraged loan futures (LVE) which reference the same benchmark as the TRS will offer market participants another avenue to gain macro leveraged loan exposure. As we have seen with other Credit Index Futures, we expect the product to add liquidity to the existing products and to further client engagement with both loans and Futures. Our introduction to Credit Index Futures can be found in [An Introduction to Credit Index Futures](#) and we reproduce portions of it below with reference to the new product.

Figure 6: ETF Leveraged Loan Market Market Size



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 7: Loan, ETF and TRS Volumes



Source: J.P. Morgan, Bloomberg Finance L.P., DTCC, LSTA

Introducing Credit Index Futures

Uses

The case for credit futures is that they provide flexible and scalable instruments for managing corporate risk exposures. The standardised nature of the instrument, along with it being exchange traded and centrally cleared, means that it has appeal to a broad array of market participants and can provide cost-effective access to corporate benchmarks. Typical use cases are:

- **Liquidity and cash management.** When large fund inflows occur, it can be efficient to take exposure to the benchmark index returns via futures which are unfunded while maintaining cash and searching for the right assets to buy.
- **Hedging purposes.** For a fund pursuing an active asset selection aimed at outperforming the market, futures can be used to temporarily reduce portfolio market exposure.
- **Portfolio overlay and beta adjustment.** Futures can be used to increase or reduce market exposure in a portfolio.
- **Tactical and relative value trades.** Credit Index Futures typically reference the same or similar markets to ETFs, TRS and CDS indices. As such, investors can trade relative value between these instruments to take advantage of pricing discrepancies.

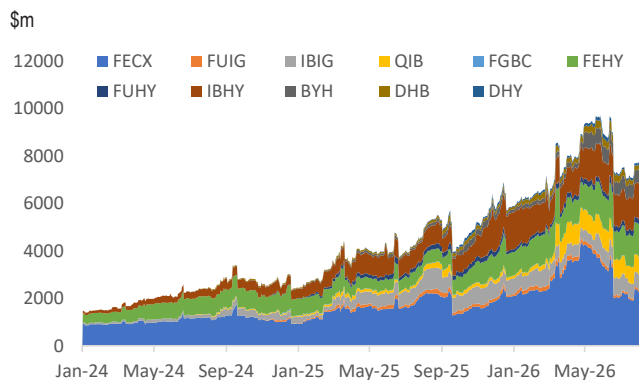
Exchange Offerings

Credit Index Futures are currently offered by three exchanges – Eurex, Cboe and CME. All three have futures referencing USD Investment Grade and High Yield indices. In addition, Eurex offers futures on Euro and Sterling Investment Grade and Euro High Yield indices. CME has also launched duration-hedged USD Investment Grade and High Yield products, which give investors exposure to the credit spread component of the total return while hedging the rates component as well as maturity-bucketed futures referencing the 1-5, 5-10 and 10+ segment of the USD IG market.

Volumes and Open Interest

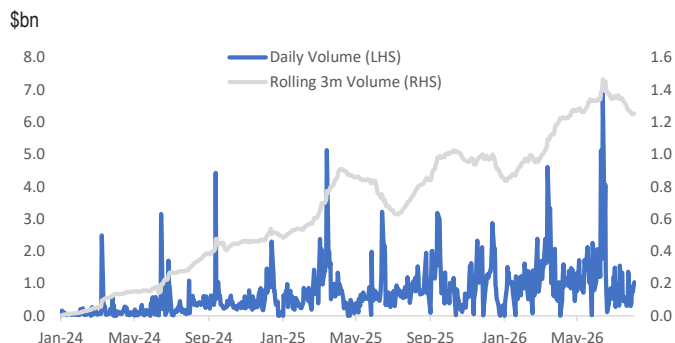
Volumes and open interest in credit futures have grown sharply over the past few years, with open interest recently passing \$9bn and average daily volumes of close to \$1.3bn over the past few months (Figure 8 and Figure 9). Open interest had fallen recently, primarily in European Investment Grade, but volumes have held up strongly.

Figure 8: Open interest across Credit futures



Source: J.P. Morgan.

Figure 9: Traded volumes



Source: J.P. Morgan.

By open interest, Eurex and Cboe are the largest futures with EUR IG futures offered by EUREX currently close to \$2.3bn in outstanding notional (Table 2). From a daily traded volume perspective, however, most products exhibit similar liquidity, trading around \$100-200mm on a daily basis in the first half of 2026.

Table 2: Credit Futures Volume and Open Interest by Ticker

\$ million

Future Ticker	Exchange	Bloomberg Ticker	Currency	Market Segment	Year-End Open Interest			Average Daily Volume		
					2026 YtD	2025	2024	2026 YtD	2025	2024
FECX	Eurex	LXY	EUR	Investment Grade	2,289	2,098	939	251	161	139
FUIG	Eurex	BBE	USD		87	80	14	52	111	15
IBIG	CBOE	IHB	USD		648	661	279	112	76	30
QIB	CME	IQB	USD		619	108	36	152	109	32
FGBC	Eurex	XZR	GBP	High Yield	9	7	14	4	6	15
FEHY	Eurex	AHW	EUR		1,377	990	700	106	88	61
FUHY	Eurex	BBL	USD		233	125	10	26	86	10
IBHY	CBOE	IBY	USD		1,561	1,556	406	156	97	37
BYH	CME	HYB	USD	IG - Hedged	563	121	22	75	62	22
DHB	CME	DHB	USD		200	103	28	244	68	46
DHY	CME	DHY	USD		104	29	-	81	41	-

Source: J.P. Morgan.

Benchmarks and ETFs

Leveraged loan Futures, ETFs and TRS all give access to the USD leveraged loan market, but the exposure of each product is different so it is worth considering what these differences are.

There are two benchmarks that underly the access products, S&P UBS USD Liquid Leveraged Loan (LLL) Index, which is the benchmark for the Future, the TRS and the SRLN ETF and the Morningstar LSTA US Leveraged Loan 100 which is the benchmark for the BKLN ETF as well as the secondary benchmark for the SRLN ETF.

S&P UBS USD Liquid Leveraged Loan Index [link](#)

The S&P UBS USD Liquid Leveraged Loan Index measures the performance of liquid USD-denominated leveraged loans and is a subset of the S&P UBS USD Broad Leveraged Loan Index. The index comprises sub-investment grade and unrated loans with a minimum facility size of \$500 million. Loans must have a minimum initial time to maturity of one year at issuance, with constituents remaining in the index until maturity. The index has a monthly rebalance schedule and targets 100 loans selected based on their liquidity score.

Morningstar LSTA US Leveraged Loan 100 Index [link](#)

The Morningstar LSTA US Leveraged Loan 100 Index is designed to measure the performance of the 100 largest facilities in the US leveraged loan market. Index constituents are market-value weighted, subject to a single loan facility weight cap of 1.9%. The parent index, the Morningstar LSTA US Leveraged Loan Index, is the starting universe. The index is reconstituted every six months, but has a weekly rebalance schedule to allow new loans to enter if existing loans have been paid down.

We highlight the main characteristics of each benchmark in Table 3.

The naming convention of the loan indices can be somewhat confusing, so it is worth noting the origins of the various indices. The Morningstar LSTA loan indices were originally part of S&P but were sold to Morningstar in 2022 when their name changed; their Bloomberg ticker still includes a “SP” prefix.

In 2024, S&P announced that had bought the previously named Credit Suisse Loan Indices and would be rebranding these as the S&P UBS Loan Indices. In Bloomberg, these tickers tend to have a “CS” prefix.

Separately, the S&P UBS USD Loan indices, not to be confused with the S&P UBS Loan indices above, were originally part of Markit iBoxx and were initially named Markit iBoxx Loan indices. In October 2025 they were renamed as the S&P UBS USD Loan indices. These are the benchmarks for the Futures. These usually have an “IB” prefix in Bloomberg.

Table 3: Benchmark Characteristics

Benchmark Name	S&P UBS USD Liquid Leveraged Loan	Morningstar LSTA US Leveraged Loan 100
Bloomberg Ticker	IBXLLTR	SPBDLL
Rebalance	Monthly	Semi-Annual Reconstitution with Weekly Rebalance
Loan Type	USD Term Loans	USD Term Loans
Minimum Size	\$500m	\$50m
Liquidity Constraint	Liquidity Ranking	100 Largest Facilities
Rating	Sub IG	-
Initial Time to Maturity	>1y	>1y
Initial Spread		Base+125bp
Remaining Time to Maturity	-	-
Size Caps		
Facility		2.0% 1.9%
Issuer		5.0%
Industry		15.0%
Start Date	31/03/2008	28/12/2001
Previous Name	iBoxx USD LiquidLeveraged Loan	S&P/LSTA U.S. Leveraged Loan 100

Source: J.P. Morgan.

Having looked at the benchmarks, we next turn to the ETFs themselves. Replication in the loan ETF market is more difficult than in the bond market as not all loans may be easily available. The ETFs aim to replicate their benchmarks, but typically don't have easy access to the bonds underlying the benchmark as such their tracking error is often higher than what we might see for more liquid bonds and equities.

Invesco Senior Loan ETF [link](#)

The Invesco Senior Loan ETF is based on the Morningstar LSTA US Leveraged Loan 100 Index. The Fund will normally invest at least 80% of its total assets in the component securities that comprise the Index. The Index is designed to track the market-weighted performance of the largest institutional leveraged loans based on market weightings, spreads and interest payments. The Fund does not purchase all of the securities in the Index; instead, the Fund utilizes a "sampling" methodology to seek to achieve its investment objective. The Fund and the Index are rebalanced and reconstituted bi-annually, in June and December.

State Street Blackstone Senior Loan ETF [link](#)

The State Street Blackstone Senior Loan ETF seeks to provide current income consistent with the preservation of capital. In pursuing its investment objective, the Fund seeks to outperform the Markit iBoxx USD Liquid Leveraged Loan Index (the "Primary Index") and the Morningstar LSTA U.S. Leveraged Loan 100 Index (the "Secondary Index") by normally investing at least 80% of its net assets (plus any borrowings for investment purposes) in Senior Loans. For purposes of this 80% test, "Senior Loans" are first lien senior secured floating rate bank loans.

Composition

To understand the exposure that the Future provides, we compare the underlying reference index to the broader market as well as to that of the ETFs in terms of rating, maturity, sector and size. As noted earlier, the benchmark, or reference index, for the TRS and Future is the S&P UBS USD Liquid Leveraged Loan Index. The index is a

subset of the Broad Leveraged Loan Index, which tracks the broader leveraged loan market. The ETFs (BKLN and SRLN) track the Morningstar LSTA US Leveraged Loan 100; we do not have a composition for this index, but the more important composition is that of the ETFs themselves. The comparison is detailed in Table 4.

Issuers and Issues

The Liquid Leveraged Loan index is by construction more concentrated than the broader loan benchmark as well as the ETFs. The LLL index has 100 loans from 88 issuers compared to 1,458 loans from 1,247 issuers in the broader market and 164 and 737 loans from BKLN and SRLN respectively. We use the J.P. Morgan Leveraged Loan Index ([JLLILLI](#)) to represent the broader loan index. The more limited number of issuers in the LLL index results in it giving more equal exposure to each underlying loan.

The largest 10 issuers in the LLL index make up 23% of the composition compared to just 5% of the broader index, but the ETFs also have high exposure to these names with the top 10 names contributing 15% of BKLN and 20% of SRLN composition weight. The largest issuer in the LLL index, TransDigm is also the largest issuer in the broader benchmark and ETFs.

Table 4: Benchmark and ETF Characteristics

Ticker	IBXXLLTR	JLLILLI	BKLN	SRLN
ETF Name			Invesco Senior Loan ETF	State Street Blackstone Senior Loan ETF
Benchmarks	S&P UBS USD Liquid Leveraged Loan Index	J.P. Morgan Leveraged Loan Index	Morningstar LSTA US Leveraged Loan 100	S&P UBS USD Liquid Leveraged Loan
				Morningstar LSTA US Leveraged Loan 100
Ticker			SPBDLL	IBXXLLTR
				SPBDLL
Maturity				
Average Maturity	4.38	4.76	4.71	5.23
7y+	3%	12%	15%	25%
Rating				
Average Rating	B1	B1	B1	B1
Baa	1.2%	5.9%	6.3%	3.6%
Caa	0.4%	3.6%	1.9%	2.3%
Sector				
Indus	22.7%	21.5%	12.3%	10.5%
Con-C	18.9%	19.0%	21.0%	20.2%
Tech	16.8%	17.5%	24.7%	26.1%
Companies				
Companies	1074	1247	104	526
Loans	1290	1458	164	737
Largest 10 Issuers	23%	5%	16%	20%
1	TDG (4.5%)	TDG (0.8%)	TDG (2.0%)	TDG (2.9%)
2	QUIKHO (3.1%)	WBD (0.8%)	HOLX (1.9%)	AAL (2.4%)
3	ALANT (2.0%)	CULLHO (0.6%)	WBD (1.9%)	HOLX (2.2%)
4	SEDGEW (2.0%)	QUIKHO (0.5%)	ULTI (1.7%)	TIBX (2.0%)
5	JANEST (2.0%)	HOLX (0.5%)	TIBX (1.6%)	ULTI (1.9%)

Source: J.P. Morgan.

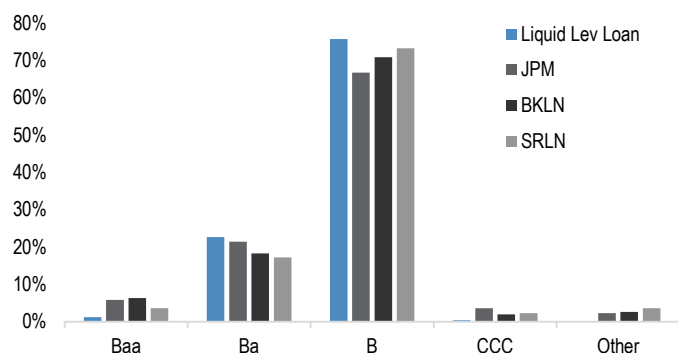
Rating

The average rating of the leveraged loan products is B1 with between 65%-75% of loans rated B. While the average ratings are similar, the Liquid Leveraged Loan index has more B and Ba rated loans than both the broader market as well as the ETFs which have both more higher as well as lower rated issues (Table 4 and Figure 10).

Maturity

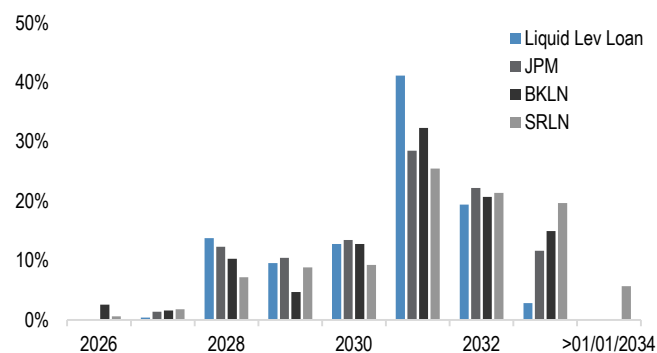
From a maturity perspective we find that the Liquid Leveraged Loan index has an average maturity of 4.4 years, lower than the broader benchmark (4.7 years), and also shorter than both BKLN (4.7y) and SRLN (5.2y). This is largely explained by the inclusion of bonds and other private placements in the ETFs which aim to replicate without exactly tracking their benchmark.

Figure 10: Exposure by Rating



Source: J.P. Morgan.

Figure 11: Exposure by Maturity

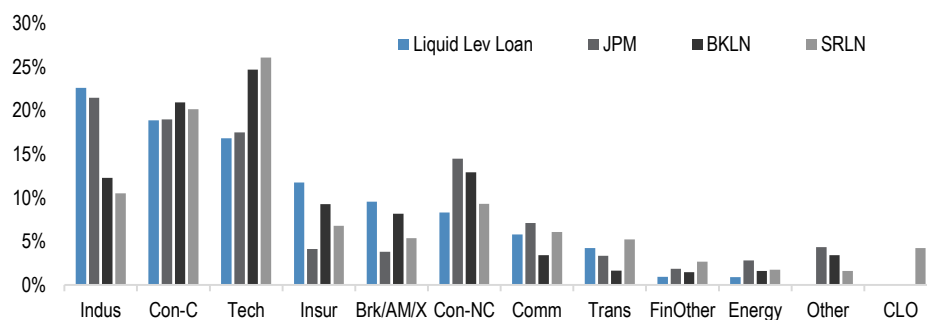


Source: J.P. Morgan.

Sectors

Sectors also vary across the products, but the top three sectors, Industrials, Consumer Cyclicals and Tech are just shy of 60% in all products. The ETFs skew more heavily towards Tech while the LLL index skews towards Industrials.

Figure 12: Exposure by Sector



Source: J.P. Morgan.

Returns and Correlations

The differing compositions in the loan access products has resulted in different returns across products. The benchmark underlying the TRS (and Future) returned 1% YtD while the benchmark underlying the ETFs Morningstar LSTA 100 has returned 1.8%. Similarly, excess returns have been negative across the board in the macro liquid products, while the broader benchmarks have eeked out positive returns (Table 5). Draw downs have also generally been more aggressive in the liquid products although relative to the magnitude of the draw down the divergence has been relatively limited. For example the LLL had a 21.3% drawdown in the COVID sell-off, while the J.P. Morgan Loan index was down 19.5%.

In the table below we also present other USD high yield access products, namely HYGH representing the USD High Yield market and CDX.NA.HY to reflect the Synthetic CDS market. We also show the total return (funded) and excess return (unfunded net of RFR) for different indices and products.

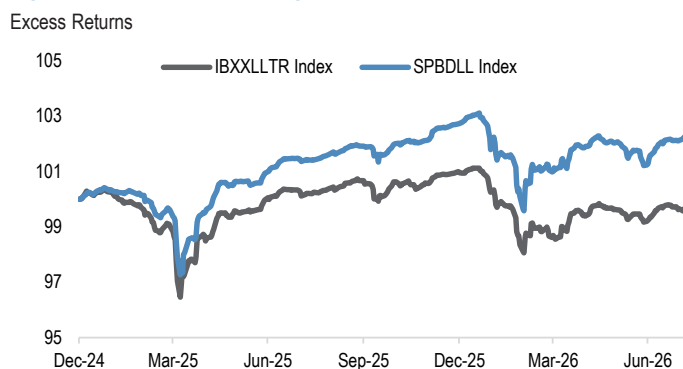
Table 5: Annualised Returns and Draw Downs in the Leveraged Loan Products

Ticker	Name	Total Return				Excess Return				Draw Downs		
		YtD	1y	5y	10y	YtD	1y	5y	10y	03/03/2026	04/04/2025	23/03/2020
IBXXLLTR Index	S&P UBS USD Liquid Leveraged Loan	1.0%	7.5%	7.3%	5.7%	-1.2%	1.3%	1.9%	2.2%	-2.4%	-3.1%	-21.3%
SPBDLL Index	Morningstar LSTA US Leveraged Loan 100	1.8%	9.5%	8.9%	7.4%	-0.4%	3.2%	3.4%	3.9%	-2.4%	-2.3%	-21.2%
IBOXLTRI Index	S&P UBS USD Broad Leveraged Loan	2.0%	7.7%	8.2%	7.0%	-0.2%	1.5%	2.7%	-	-1.7%	-2.5%	-19.7%
JLLILLI Index	J.P. Morgan Leveraged Loan	2.4%	9.0%	9.2%	8.0%	0.2%	2.7%	3.7%	4.4%	-1.5%	-2.2%	-19.5%
SPBDAL Index	Morningstar LSTA US Leveraged Loan	2.3%	8.8%	8.8%	7.7%	0.1%	2.5%	3.3%	4.1%	-1.7%	-2.0%	-19.5%
CSLLLTOT Index	S&P UBS Leveraged Loan	2.4%	8.8%	8.7%	7.7%	0.1%	2.5%	3.2%	4.1%	-1.6%	-2.0%	-19.2%
BKLN US Equity	Invesco Senior Loan ETF	1.2%	9.0%	7.8%	6.0%	-1.0%	2.7%	2.3%	2.4%	-2.5%	-3.2%	-22.0%
SRLN US Equity	State Street Blackstone Senior ETF	2.0%	9.8%	6.9%	6.4%	-0.2%	3.5%	1.5%	2.9%	-2.8%	-3.8%	-19.6%
HYGH US Equity	iShares IR Hedged HY Bond ETF	4.1%	13.0%	10.1%	8.6%	1.8%	6.5%	4.6%	5.0%	0.2%	-6.2%	-22.3%
USIRHHYC Index	BlackRock IR Hedged HY Bond	3.7%	11.7%	-	-	1.4%	5.3%	-	-	0.0%	-3.9%	-
ERINCDHY Index	CDX.NA.HY 5-year	2.9%	11.7%	9.5%	7.9%	0.6%	5.3%	4.0%	4.3%	-0.5%	-3.9%	-18.0%

Source: J.P. Morgan.

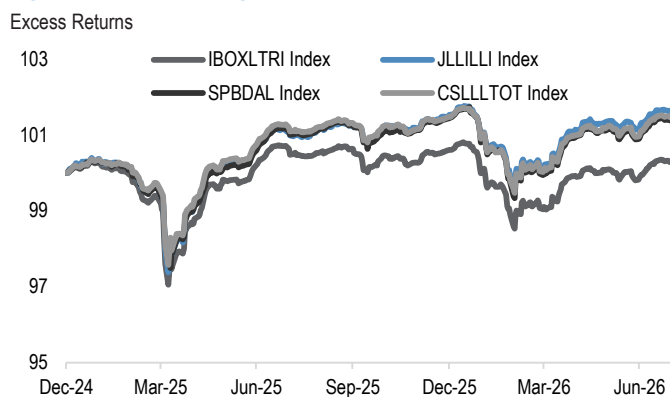
Over the past couple of years the Morningstar index (SPBDLL) has outperformed the S&P (IBXXLLTR) index which we believe is implementation-driven rather than a pure difference in underlying loan risk. The S&P liquid loan index is designed to be more executable and, in doing so, appears to embed bid/offer costs around index turnover, which can create a persistent performance headwind. For instance, a similar outperformance of the Morningstar index is visible versus BKLN ETF which is benchmarked to the Morning star index. A smaller portion of the gap is consistent with Morningstar carrying a modest tilt to longer-dated and lower-rated loans (Figure 13). When we compare the returns of the broad benchmark indices, we find that they have shown almost identical excess returns, apart from the broader S&P (formerly iBoxx) index which has a higher minimum outstanding amount (Figure 14).

Figure 13: S&P versus Morningstar Liquid Loan



Source: J.P. Morgan.

Figure 14: Broad Leveraged Loan Indices

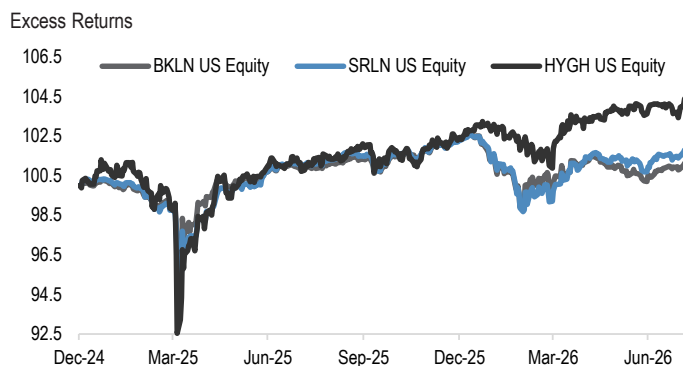


Source: J.P. Morgan.

When we move on to considering the ETF products we find that their returns have been very similar over the past couple of years, although there are periods of out- or underperformance between the two. They have also exhibited less volatility than the high yield ETF HYGH although they have underperformed this year on the back of Software exposure weakness to which they are more exposed than the high yield ETF (Figure 15).

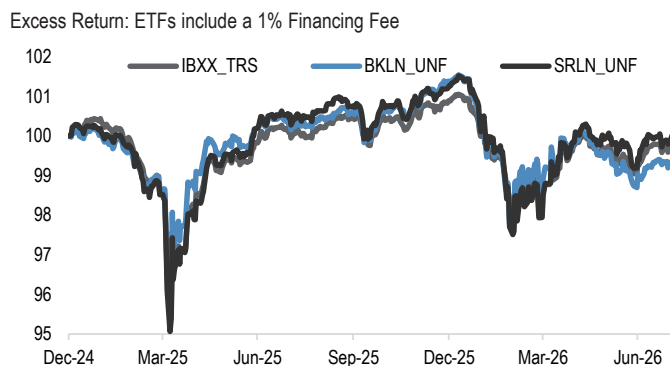
Finally, we look at the performance of the TRS assuming an investor traded the front contract and rolled it every three months to that of the ETFs with a 1% financing fee (Figure 16). Here too we see that returns and performance has been very similar throughout the period, suggesting that the choice of ETF or TRS, or Future, is likely a preference of format rather than exposure.

Figure 15: Loan versus High Yield ETFs



Source: J.P. Morgan.

Figure 16: TRS versus ETF Returns



Source: J.P. Morgan.

To conclude this section, we look at the correlation of weekly excess returns across the different loan and high yield products. The results suggest that at least on a weekly basis returns across the different indices are highly correlated although there is also diversification to be gained from trading loans versus bonds versus CDS (Table 6).

Table 6: Correlation of Weekly Excess Returns

Five year lookback

	IBXXLLTR Index	SPBDLL Index	IBXXSLLT Index	IBOXLTRI Index	SPBDAL Index	CSLLLTOT Index	BKLN US Equity	SRLN US Equity	HYGH US Equity	USIRHHYC Index	ERINCDHY Index
IBXXLLTR Index	100%										
SPBDLL Index	99%	100%									
IBXXSLLT Index	96%	97%	100%								
IBOXLTRI Index	97%	96%	100%	100%							
SPBDAL Index	97%	97%	99%	100%	100%						
CSLLLTOT Index	97%	96%	100%	100%	100%	100%					
BKLN US Equity	95%	95%	82%	91%	91%	91%	100%				
SRLN US Equity	91%	91%	86%	93%	93%	93%	93%	100%			
HYGH US Equity	72%	69%	61%	69%	69%	69%	78%	77%	100%		
USIRHHYC Index	65%	62%	62%	62%	60%	62%	82%	79%	98%	100%	
ERINCDHY Index	53%	50%	57%	52%	51%	51%	61%	64%	87%	92%	100%

Source: J.P. Morgan.

Contract mechanics

Credit futures are cash-settled products traded on an underlying total return corporate index. The cash-settled nature of the product means that at expiry, the investor will pay (or receive) the difference between the initial entry price and the final price on settlement, determined by the exchange. Typically, however, market participants will either roll or unwind their position prior to the futures expiry such that few futures are likely to settle at the final price. Daily margining of the product means that there is unlikely to be a large payment on exit as the total return over the period includes the daily mark-to-market valuation.

Unlike single bond futures, which settle on a cash price of the bond (with a factor adjustment), credit index futures settle on a total return index. The daily return of the underlying index is the aggregate price return of all the underlying bonds in the index, plus the daily accrued from each bond. It is, in essence, a dirty price index that includes any coupons paid by the underlying bonds. Over time, the index accrues positively, reflecting the total return of the bonds in the index.

For example, the loan future offered by CME, LVE (Bloomberg Ticker: LVE), references the S&P UBS USD Leveraged Loan Index (IBXXLLTR). This underlying index includes all USD-denominated leveraged loans that meet the inclusion criteria set by S&P. The future price reflects the level of the index plus funding to the term of the future. The upward trend in the index differs from the price of a comparable ETF as coupons/dividends are not paid out of the index but rather accumulated (Figure 17).

Table 7: Contract Description

PRODUCT	S&P UBS USD Liquid Leveraged Loan Index futures
CONTRACT UNIT	\$100 x index
BLOOMBERG TICKER	LVE index
SETTLEMENT METHOD	Financially settled
PRICING QUOTATION	Index price
LISTING SCHEDULE	Nearest three quarterly months (Mar, Jun, Sep, Dec)
TERMINATION OF TRADING	Trading terminates on the business day before the third Wednesday of the contract month
BLOCK RULES	50 contracts with a 15-minute reporting window; derived block eligible
TRADING AND CLEARING HOURS	Globex: Sunday through Friday 5:00 p.m. to 4:00 p.m. Central time (CT) ClearPort: Sunday through Friday 5:00 p.m. to 4:00 p.m. CT

Source: CME

Expiry

The three exchanges offer quarterly expiries – March, June, September and December – with most liquidity focused on the front contract. This means that in August, the September contract would be the most liquid across the available options. As expiry approaches, investors will typically roll onto the next available contract (Figure 18) in the two weeks prior to expiry. The roll expiry date does differ between the exchanges with Eurex settling on the third Friday of the expiry month, Cboe settling on the first business day of the expiry month and CME settling on the third Wednesday of the expiry month. In the appendix we detail the historical performance and cost of the quarterly rolls.

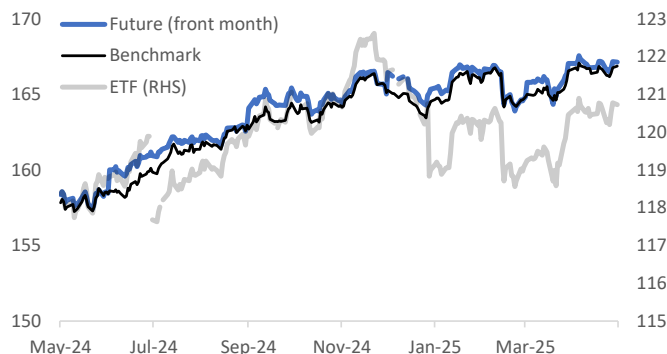
Trading Hours

One of the attractions of futures is that they provide almost round-the-clock trading, thereby extending the hours at which market participants can adjust their exposure to the underlying market. Eurex hours are 7.50am to 10.00pm CET, Cboe and CME hours are

5.00pm the previous day to 4.00pm Chicago time.

Figure 17: Euro IG Future Contract and Benchmark

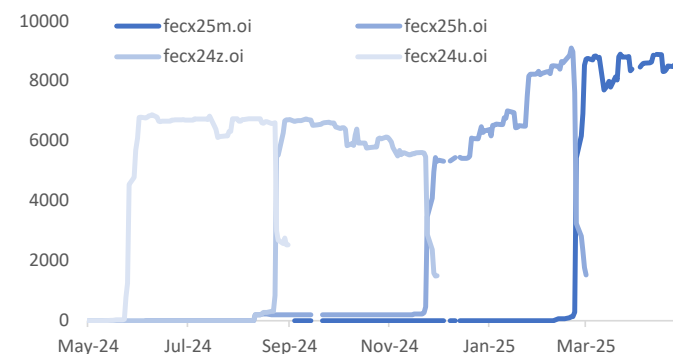
LHS: Credit Future and Benchmark, RHS: Comparable ETF Price



Source: J.P. Morgan, EUREX, Bloomberg Finance L.P.

Figure 18: FECX Open Interest

Number of contracts



Source: Eurex

Contract Sizing and Multipliers

Contracts trade with a contract multiplier that gives the investor a notional exposure to the underlying index. The leveraged loan future has a contract multiplier of 100. Suppose the traded future has a price of 233.01; this means that one contract will have market exposure of \$23,301. This is a market value exposure rather than notional so is equivalent to Bond Price * Notional for a single bond. An investor who wants to have \$100m of market exposure would therefore buy 4,291 contracts ($= \$100m / (233.01 * 100)$).

Sizing and PnL Example

A credit PM gets an inflow of \$50mm in early January and is looking for a quick and efficient way to gain all-in yield (spread and interest rate) exposure to the broad USD IG market. The PM can do so by purchasing CME LVE futures while awaiting new issuance / supporting secondary conditions to invest cash. The PnL of the trade will track the total return performance of the leveraged loan benchmark (in this case as represented by the IBXXLLTR Index) less financing costs, which could be offset by putting the cash into a money market account.

Let's assume the inflow arrives on 8 January 2027; the liquid / nearest future contract is the March Expiry (LEVH27). Assume Bid/Offer is 232.05 / 233.01. For reference, the previous EOD close of the NAV for IBXXLLTR index was 231.4.

The contract multiplier is 100 so a single contract exposure / value is \$23,301 ($= 233.01 * 100$). To get \$50mm exposure (off the offer side, but can also use mid or bid), the PM would look to purchase 2,145 contracts ($= \$50mm / \$23,301$). The trade exposure is \$50mm, but only Initial Margin changes hands initially. Given an initial margin is 5%, the PM posts \$2,500,000 at trade initiation. Margins will often be lower than this as exchanges will often give offsets for offsetting trades, e.g. on a rates CME-cleared product, the PM could get up to 75% offset. There are additional fees paid to the exchange, but these are typically small.

As the market moves, the Variation Margin will be paid/received and calculated as (Daily Settlement Price – Entry Price) x # of contracts. As the contract approaches expiry, the futures price will converge to the underlying index level (aka spot). If the PM

holds the futures to the settlement on Wednesday, 17 March (last trading date would be Tuesday 16 March), the cash settlement will be based on the NAV index fixing, for that day at 4pm ET. For example, IBXXLLTR Index NAV on 17 March was 234.86, so the total P&L would be \$396,825 $(=(234.86 - 233.01) \times 2,145 \times 100)$ or 0.80% appreciation from Trade Date on the exposure of \$50mm.

The difference between this return and the return for the benchmark itself will be roughly equal to financing the trade at SOFR (3.65%) for 70 days.

If the PM were to unwind the position prior to expiry, the PnL will be calculated in a similar fashion, but using exit price / bid quote of the contract. Alternatively, the PM may choose to keep the exposure and roll the position (by selling the “expiring” near-month futures and purchase far-month futures), typically several days prior to the contract’s last trading day to avoid PIN risk of the underlying index and ensure certainty of the trading levels, roll and financing costs.

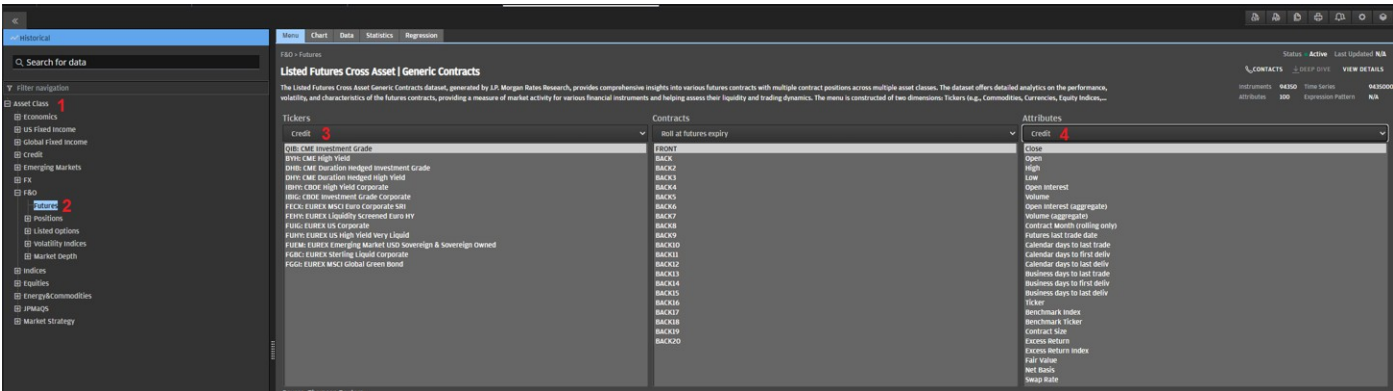
Credit Futures Data in DataQuery

To facilitate transparency in this corner of the market, we have added a rich historical dataset for Credit futures, which is accessible via [DataQuery](#). The first step to access the data is to navigate DataQuery's option menu selecting **Asset Class**→ **Credit** → **Tradeable Indices**→ **Futures**. Figure 19 shows how to navigate to the dataset on the DataQuery menu.

In addition to exchange-related data such as prices, open interest and volumes, we also provide enriched data such as future fair value, discount/premium to NAV and excess returns to help market participants analyse and understand this part of the market.

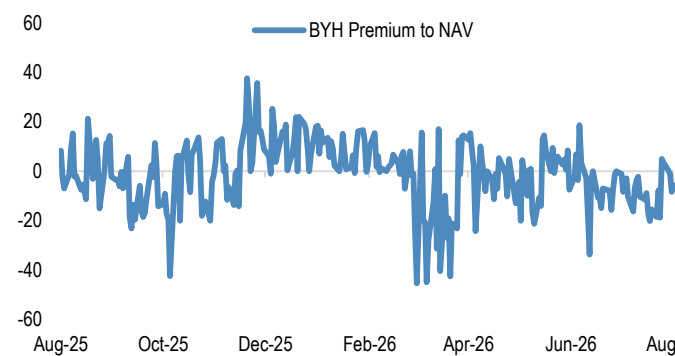
Figure 19: Credit Futures menu on DataQuery

Numbers indicate the menu navigation steps.



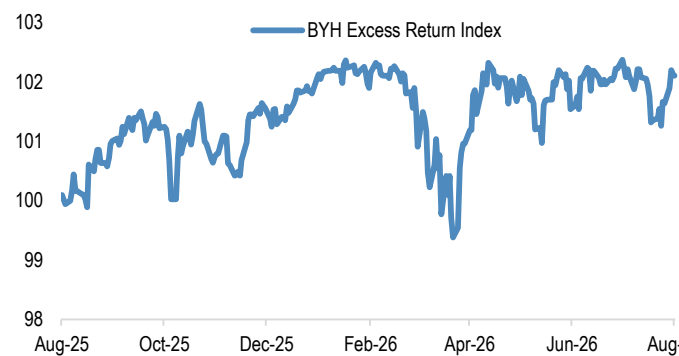
Source: J.P. Morgan, DataQuery

Figure 20: Future Premium-to-NAV



Source: J.P. Morgan.

Figure 21: Future Excess Return Index

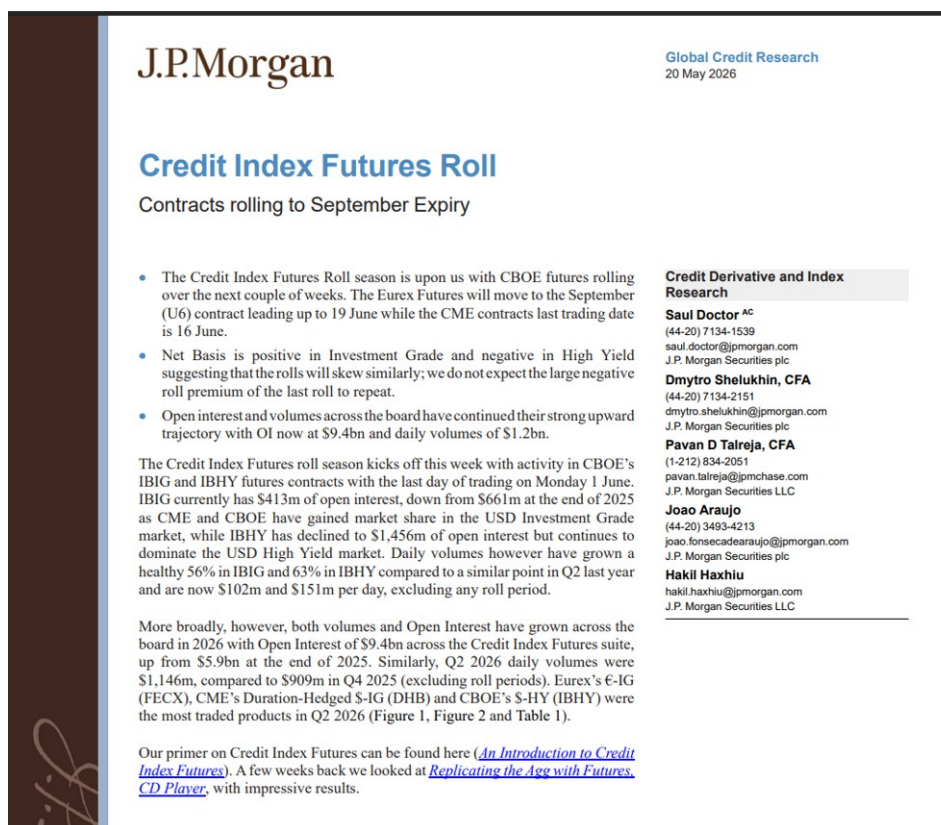


Source: J.P. Morgan.

Quarterly Roll Notes

Every three months the credit index futures roll onto the next quarterly contract. To help market participants understand the upcoming roll and how to trade it, we produce a quarterly update looking at positioning and premiums leading into each roll.

Figure 22: Credit Index Futures Quarterly Roll



Source: J.P. Morgan.

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Explanation of Credit Research Valuation Methodology, Ratings and Risk to Ratings:

J.P. Morgan uses a bond-level rating system that incorporates valuations (relative value) and our fundamental view on the security. Our fundamental credit view of an issuer is based on the company's underlying credit trends, overall creditworthiness and our opinion on whether the issuer will be able to service its debt obligations when they become due and payable. We analyze, among other things, the company's cash flow capacity and trends and standard credit ratios, such as gross and net leverage, interest coverage and liquidity ratios. We also analyze profitability, capitalization and asset quality, among other variables, when assessing financials. Analysts also rate the issuer, based on the rating of the benchmark or representative security. Unless we specify a different recommendation for the company’s individual securities, an issuer recommendation applies to all of the bonds at the same level of the issuer’s capital structure. We may also rate certain loans and preferred securities, as applicable. This report also sets out within it the material underlying assumptions used. We use the following ratings for bonds (issues), issuers, loans, and preferred securities: Overweight (over the next three months, the recommended risk position is expected to outperform the relevant index, sector, or benchmark); Neutral (over the next three months, the recommended risk position is expected to perform in line with the relevant index, sector, or benchmark); and Underweight (over the next three months, the recommended risk position is expected to underperform the relevant index, sector, or benchmark). For CDS, we use the following rating system: Long Risk (over the next three months, the credit return on the recommended position is expected to exceed the relevant index, sector or benchmark); Neutral (over the next three months, the credit return on the recommended position is expected to match the relevant index, sector or benchmark); and Short Risk (over the next three months, the credit return on the recommended position is expected to underperform the relevant index, sector or benchmark).

J.P. Morgan Credit Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
Global Credit Research Universe*	26%	58%	16%
IB clients**	84%	84%	89%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "Overweight," "Neutral" and "Underweight" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. The Credit Research Rating Distribution is at the issuer level. Issuers with an NR or an NC designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public

appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation

sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P. Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmpi.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpmpi.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpmpi.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss

arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised July 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 07 Aug 2026 11:11 PM BST

Disseminated 10 Aug 2026 07:00 AM BST